Xuanchen(Jason) Liu

Solutions Developer / Data Engineer

Jasonliu97424@gmail.com
https://www.linkedin.com/in/xuanchen-liu-jason/
https://xuanchen97.github.io/
647-979-5247

Work Experience

Solutions Developer

Aug 2023 - Present

Hamilton Health Science (Population Health Research Institute)

- Worked closely with study teams and finance stakeholders to understand complex clinical payment rules, clarify requirements, and translate them into reliable SQL-based data processing logic
- Identified gaps and inconsistencies in legacy payment workflows (Excel tracking and undocumented SQL), and helped standardize and reconstruct the data pipeline for improved accuracy and maintainability
- Developed and executed SSIS packages to synchronize data between SQL Server and Oracle systems, ensuring consistency across heterogeneous data sources
- Expanded and modernized the in-house clinical data management system by developing new features and refining existing components to support evolving study workflows
- Coordinated code integration and release processes using Git, including resolving merge conflicts and managing deployments across development and production environments
- Collaborated with cross-functional teams to deliver data-driven solutions in an Agile environment

Full-Stack Developer

Jan 2021 - Jul 2023

Orispark Inc.

- Designed and developed backend systems using C# (.NET) and Python (FastAPI), focusing on scalable data processing and API design
- Built RESTful APIs to handle data validation, transformation, and system integration across services
- Designed relational schemas and optimized queries to ensure data integrity and performance
- Investigated and resolved performance bottlenecks through query optimization and backend logic improvements

Technology Researcher & Developer (Co-op)

Sep 2019 - Aug 2020

World Vision Canada

- Developed internal applications using PHP and MySQL, focusing on data storage, retrieval, and reporting.
- Designed and tested database-driven features to support business operations.
- Collaborated with teams using Azure DevOps in an Agile environment.

Projects

Clinical Payment Data Pipeline (Finance Team)

Oct 2024 - Present

- Reconstructed and standardized a clinical payment data pipeline in the absence of documentation, ensuring accurate and consistent payment reporting
- Analyzed and refactored legacy SQL scripts to ensure accurate identification of payable patient visits
- Integrated data from multiple clinical systems (TrialMaster - SQL Server, DataFax - MySQL) into a unified payment workflow
- Ensured quarterly payment data consistency by validating and synchronizing visit-level data into the finance payment system
- Collaborated with finance and study teams to interpret complex business rules and resolve data inconsistencies

Participant Planning & Timeline Derivation Feature (Clinical Trials)

Oct 2025 - Apr 2026

- Designed and implemented a data-driven module to replace manual Excel-based payment tracking, modeling clinical study visit schedules and participant progression
- Built transformation logic to derive projected visit timelines and reporting readiness from partial and incomplete data using FPFV as a temporal anchor
- Modeled relational data structures and implemented dynamic data transformation logic to derive timelines based on user inputs and study configurations
- Implemented data validation and fallback strategies to ensure data integrity and transparency in cases of missing inputs
- Collaborated with stakeholders to define study planning requirements and ensure alignment with overall platform design

Core Skills

Languages: C#, SQL, Python, React, Node.js

Database: SQL Server, MySQL, Firebase

APIs: RESTful API development and integration

Tools & Platforms: Azure DevOps, GitHub, Jenkins

Education

Sheridan College

Sep 2017 - Dec 2020

Advanced Diploma Software Development and Network Engineering

Certificates

ITIL® V4 Foundation

Dec 2023

PeopleCert

Managing Big Data with MySQL

Sep 2020

Duke University | Coursera